## Q1.

Find any relative extrema of
$f(x) = x\sqrt{25 - x^2}$

State where the function is increasing or decreasing and determine points of inflection.

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## Q2.

Find the absolute maximum and minimum values of
$f(x) = \frac{x^2}{3x^3+1}$

on $[0, +\infty)$.

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## Q3.

The total cost and total revenue functions are
$C(x) = \frac{1}{3}x^3 - 8x^2 + 70x + 300$
$R(x) = x(200 - 4x)$

a) Write the profit function $P(x)$ and determine the level of output that maximizes profit.

b) Verify that $MR = MC$ at that point.

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## Q4.

A rectangular field is enclosed using $800$ meters of fencing on three sides, with one side along a river.

Find the dimensions that maximize the area.